

IMCA 221
Programming for Artists
Winter 2025

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**Class is on Zoom even in the classroom
for sharing, find the details on Moodle**

<https://moodle.concordia.ca/>

Download the slides!

Final projects - November 25

Prototype - November 11

- If you don't pay your fees you can't reserve rooms to present, **PAY YOUR FEES!!!!**

- Decide if you are going to work alone or in teams, this will determine our critique length

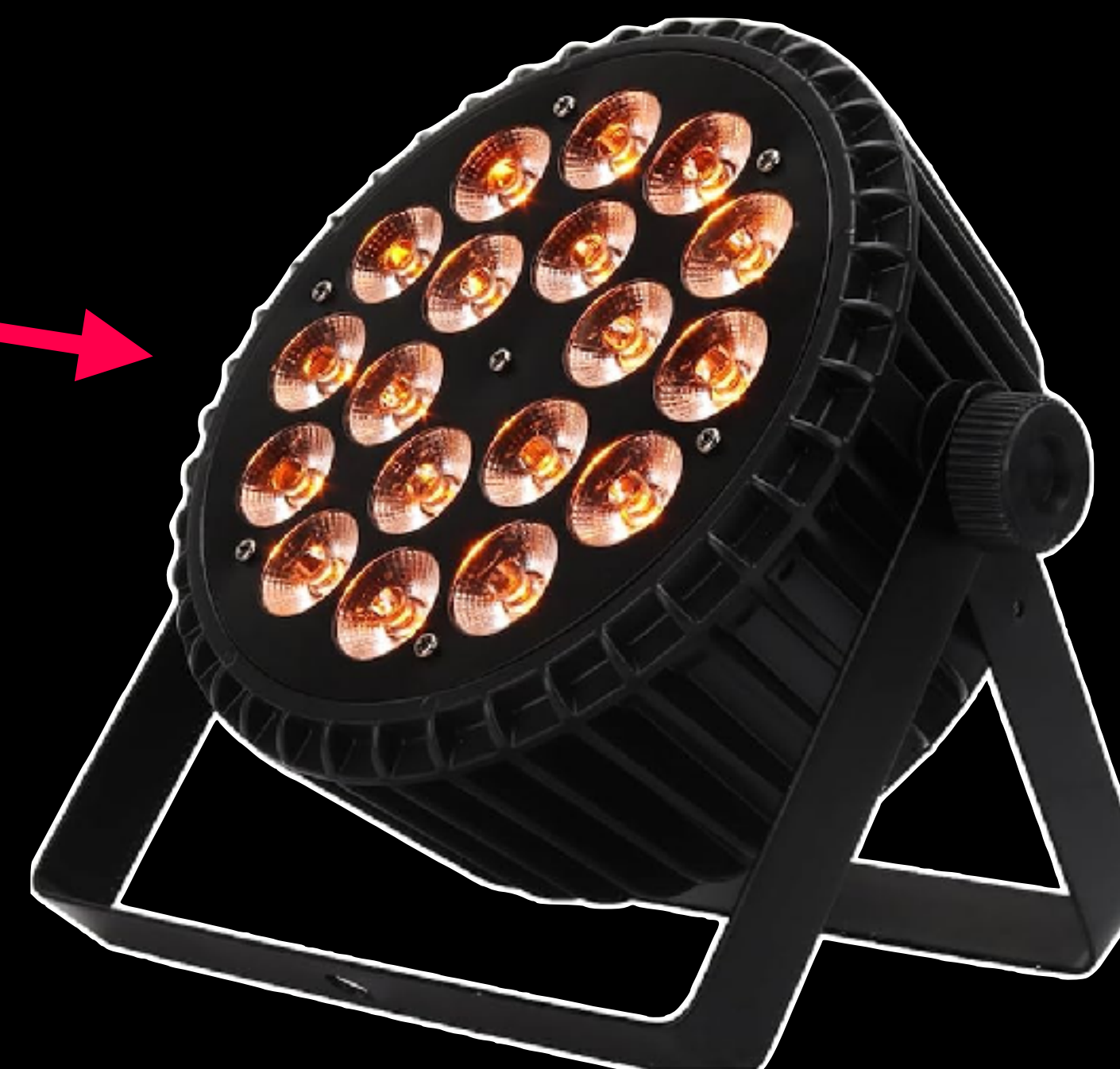
- Sign up here (times are not final)

<https://docs.google.com/spreadsheets/d/1tYaLwQvdHKLPFb9jIBjtOkV2yulljyn--IB8jXvJ52s/edit?gid=0#gid=0>

Prototype - November 11

Prototype should include:

- A working idea to show the class
- A written timeline of work to be completed (What will happen, when
- List of equipment needed for your final (things to make, cords, adapters, projectors, mounting supplies etc)
- 2 or 3 questions to ask the class during critique session



Communication

Arnet : Hardware / protocol for communicating with DMX
(this example using Beam): <https://www.youtube.com/watch?v=IEW-Wy-7R34>

MQTT: Internet-based protocol for communicating with
devices over channels

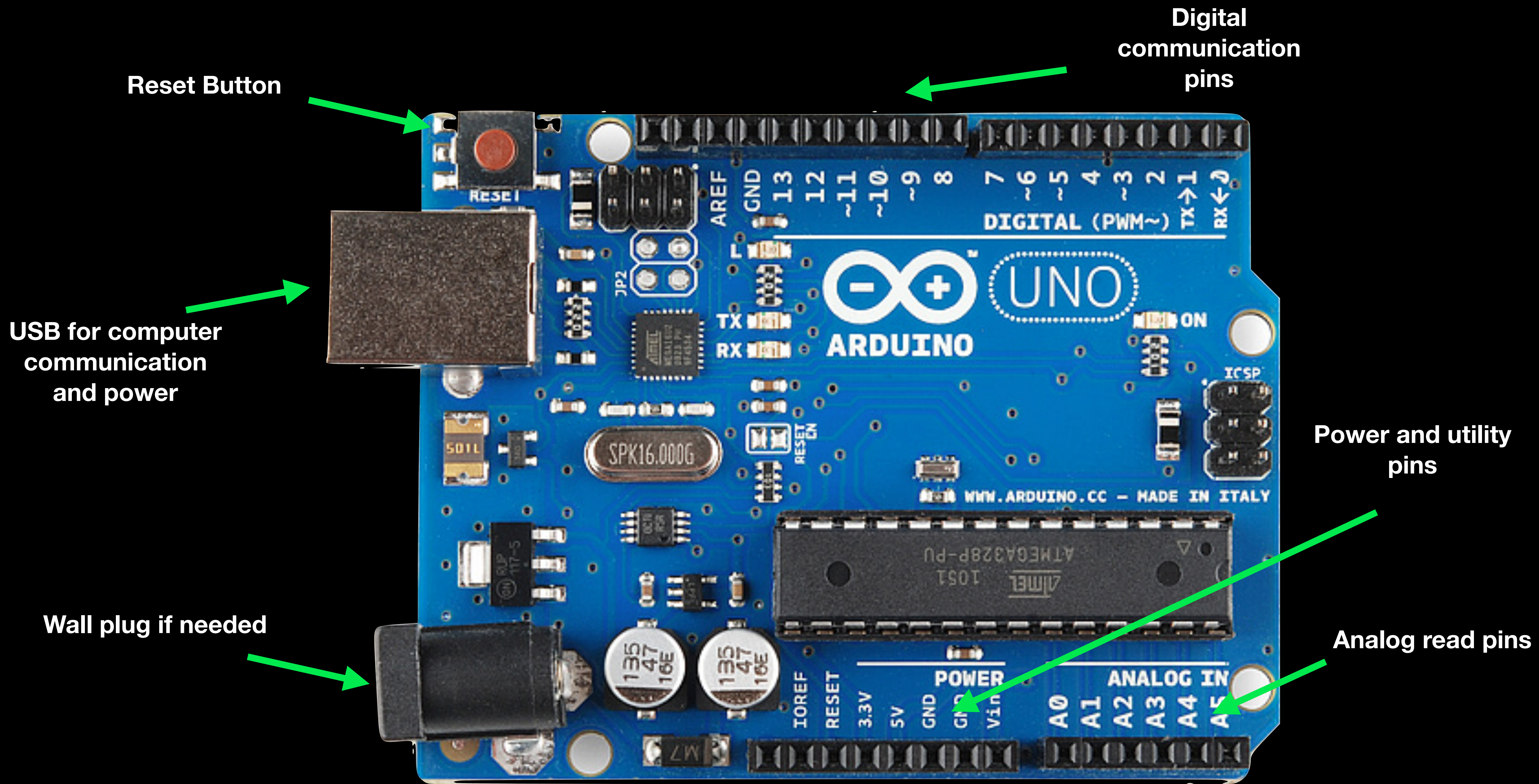
<https://docs.arduino.cc/tutorials/uno-wifi-rev2/uno-wifi-r2-mqtt-device-to-device/>

OSC: UDP communication

<https://www.youtube.com/watch?v=MFbeizEtU7Y>

Arduino
Micro controller
platform for
controlling
physical things





Serial

We will be using a serial communication protocol to communicate over USB to the patch. You can send any message over Serial and decode it at the end.



Inputs

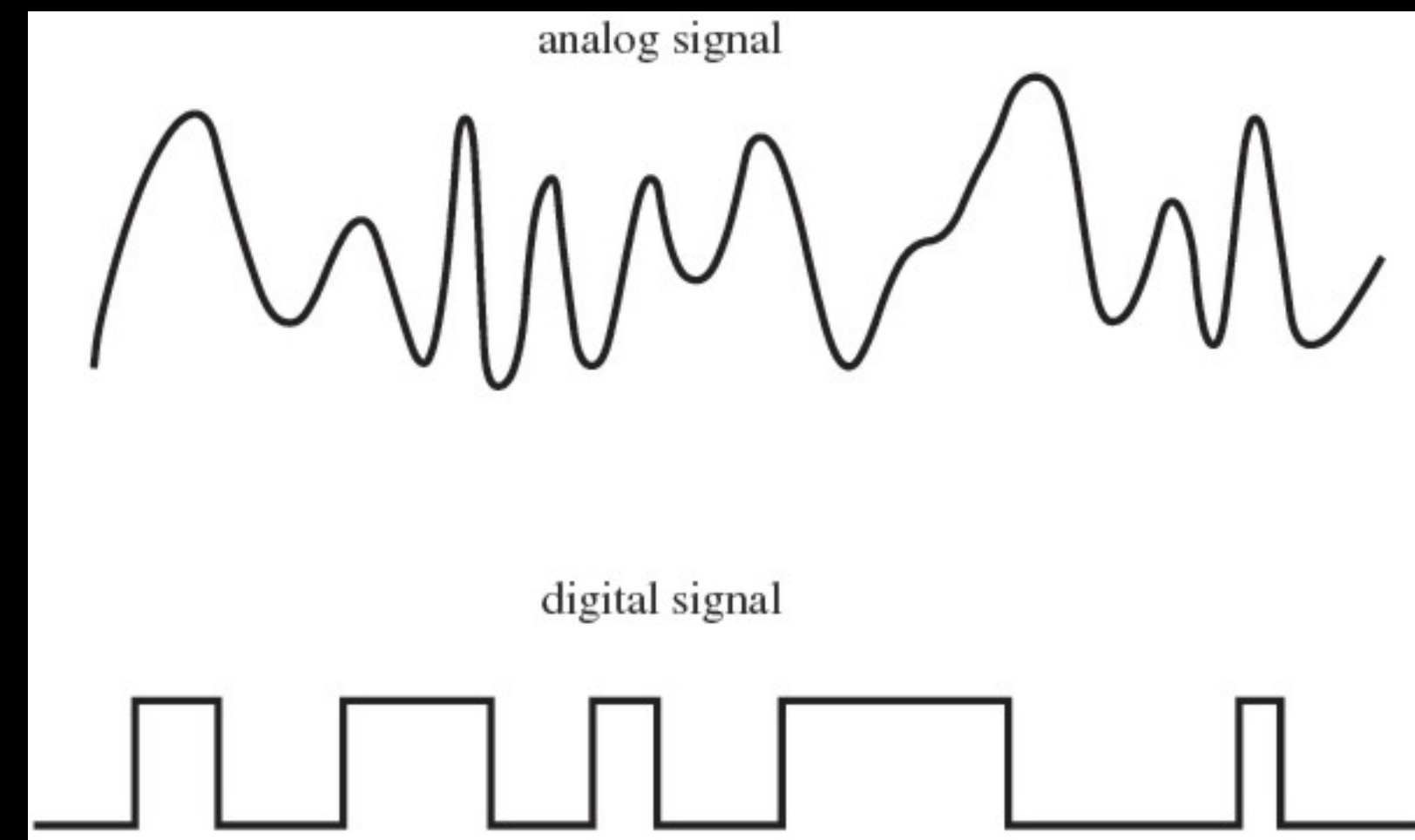
- We will be programming an Arduino to read sensor input.
- The sensor input will be sent over serial to Maxmsp
- Maxmsp will receive that data in the form of a stream of numbers we can use for anything!

For now, this is 1 way communication.



Signals:

- Analog signals can be a range of numbers.
- Digital signals are a package of 0 or 1s.



Analog Sensors

- Provide a range of numbers according to an input.
- Light sensors (LDR)
- Pressure sensors (FSR)
- Flex Sensors
- Knobs



Buttons (normal)

- Push buttons

- Switches

- Tilt Switch

(changes as you turn it)



Buttons (weird)

- You can make buttons out of anything conductive

Great resource:

<https://>

www.kobakant.at/

[DIY/](https://www.kobakant.at/DIY/)



Range Finders & Accelerometers

- Range finders indicates a distance between the sensor and nearest object in line of sight
- accelerometer can be used to determine approximate orientation and speed



So many other sensors!

- Heart rate, GPS,
temperature, humidity,
sound, vibration,
atmospheric, gyroscope...
check out more at:

- [https://www.adafruit.com/
category/35](https://www.adafruit.com/category/35) &

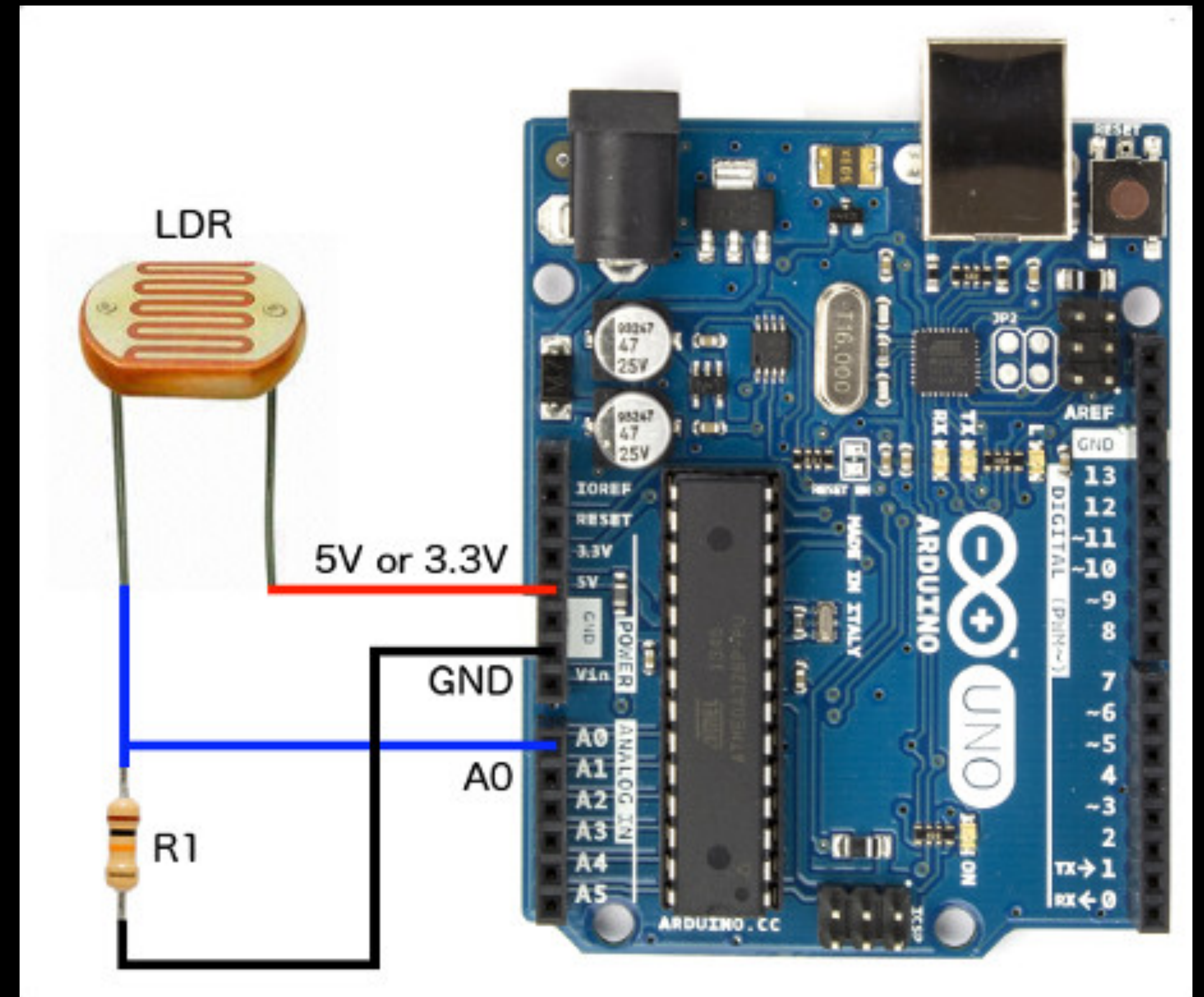
- [https://www.sparkfun.com/
categories/23](https://www.sparkfun.com/categories/23)



Hook up a sensor
(Light, Flex, Pressure,
knob)

- Red > 5v
- Black > GND
- Yellow/Blue > A0

Output: You will get 1
number that indicates
the value of the sensor



Lets open up Arduino!
[https://github.com/
LeeCyborg/Arduino-To-
MaxMSP/tree/main/
analogSensor](https://github.com/LeeCyborg/Arduino-To-MaxMSP/tree/main/analogSensor)

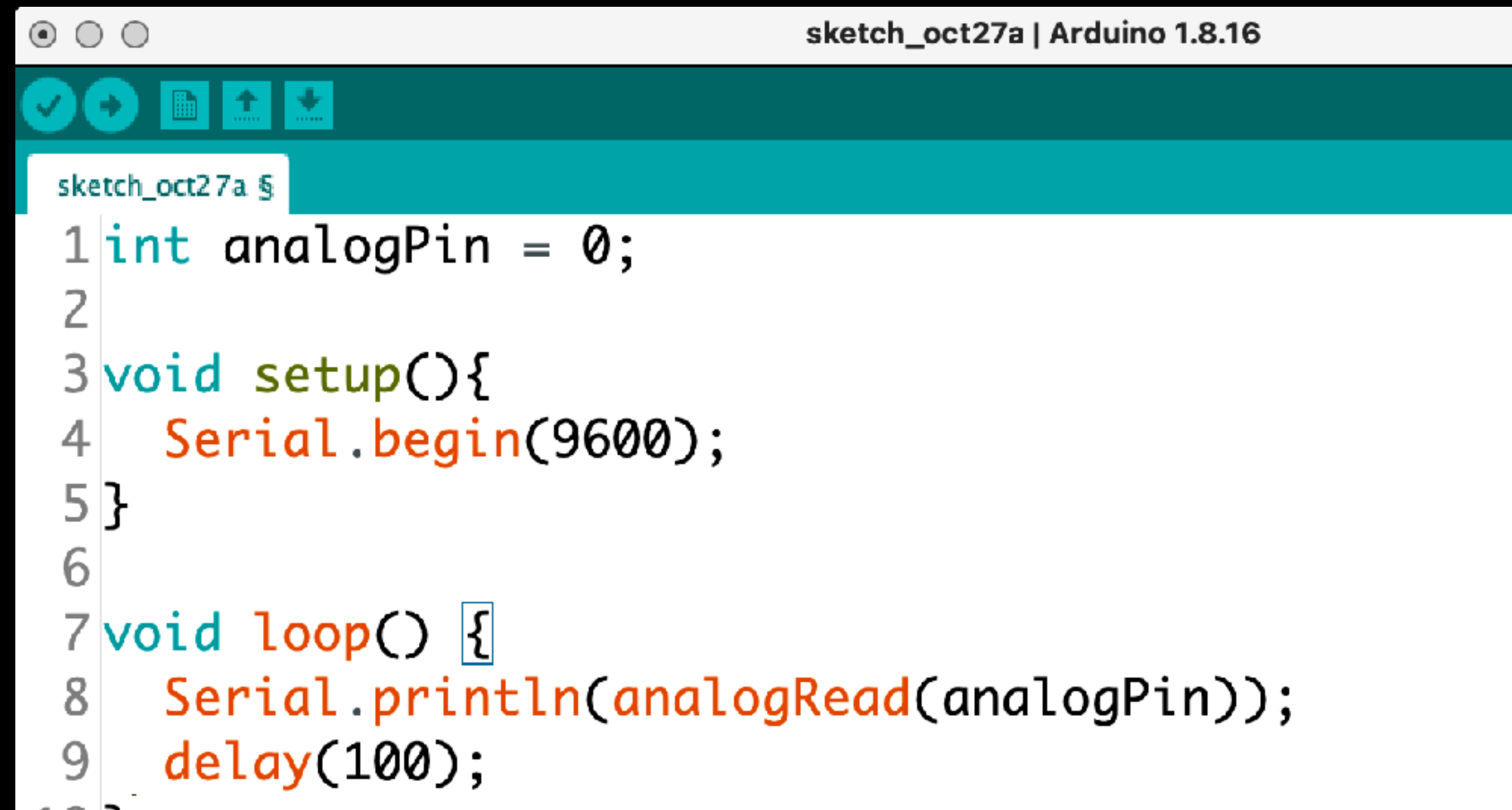
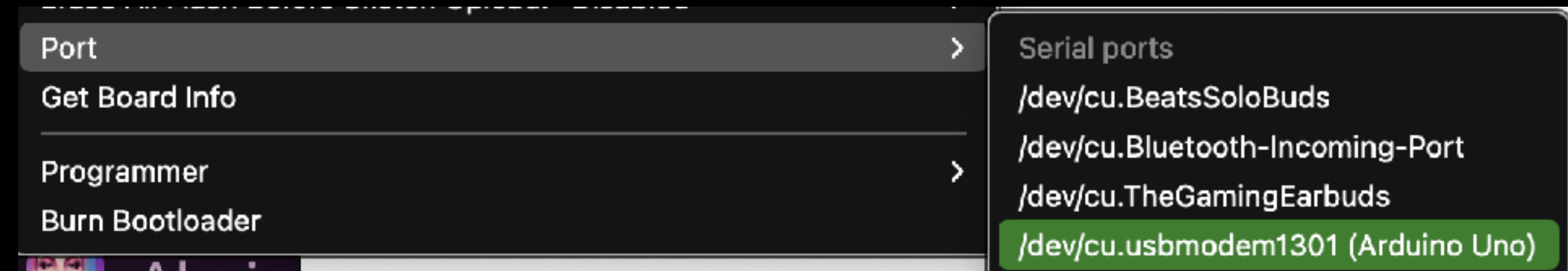
You can copy/paste the
code from here.

Click analogSensor.ino

Arduino-To-MaxMSP / analogSensor /	
LeeCyborg update baud rate for analog sensors	
Name	Last commit message
..	
CircuitDiagram.jpg	Circuit Diagrams
analogSensor.ino	update baud rate for

Open Arduino

- Plug the Arduino into the computer with USB
- Select Arduino on the port. Tools > port



This is the pin your sensor is connected to

Open a serial port between the computer and Arduino at this rate

```
int analogPin = 0;
```

```
void setup(){  
  Serial.begin(9600);  
}
```

Print it to the Serial, this is what your Max Patch is reading!

```
void loop() {  
  Serial.println(analogRead(analogPin));  
  delay(100);  
}
```

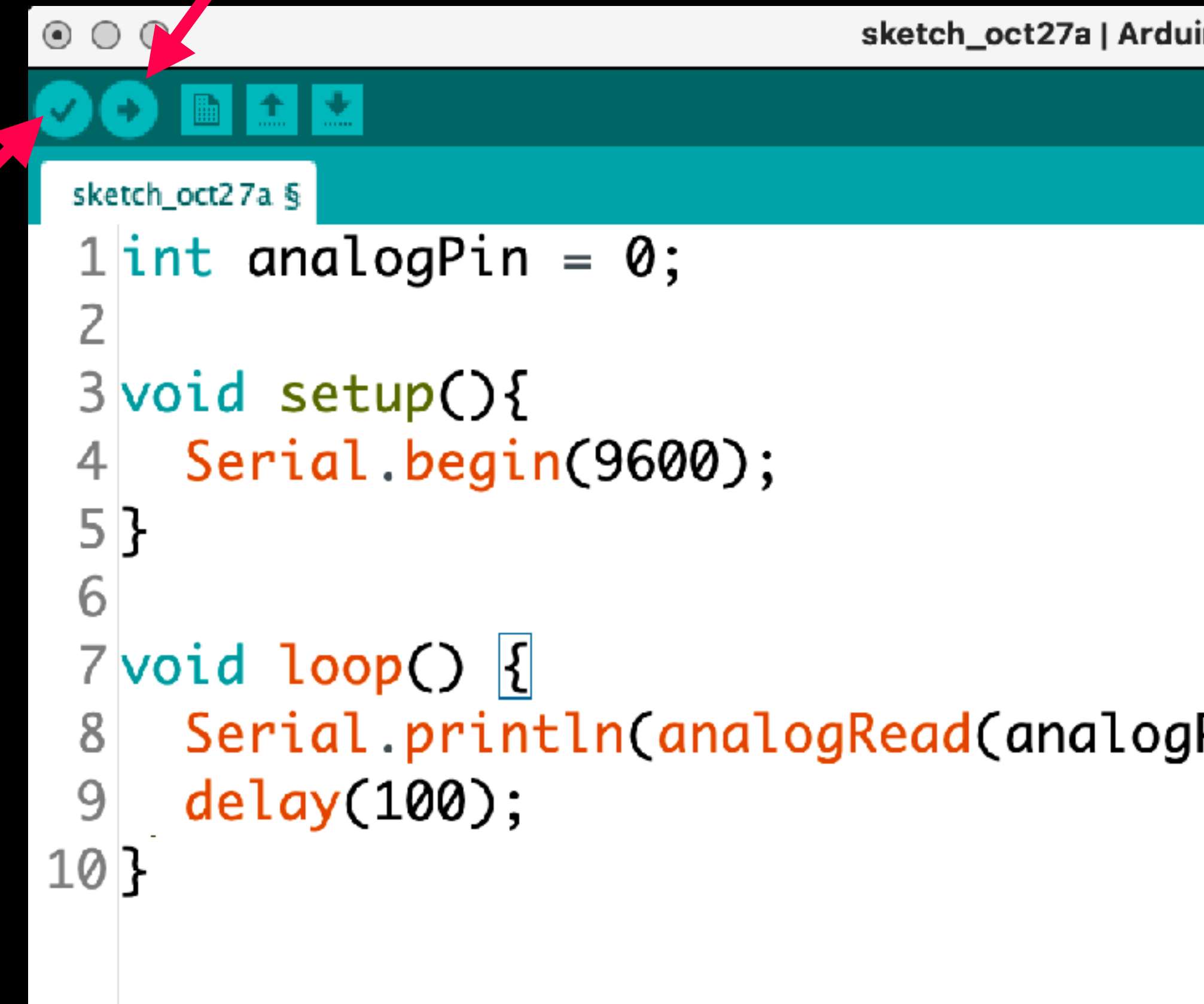
Serial.print send to serial
analogRead read sensor value
analogPin the pin to read

Upload to the Arduino

- Plug the Arduino into the computer with USB
- Click Verify to check your code has no errors
- Click upload to put the code on your board
- You should see Upload Successful on the bottom

Verify

Upload



The screenshot shows the Arduino IDE window titled "sketch_oct27a | Arduino". The top toolbar contains icons for Verify (a checkmark), Upload (an upward arrow), and others. A red arrow points to the Verify icon, and another red arrow points to the Upload icon. The main editor area contains the following C++ code:

```
sketch_oct27a $
1 int analogPin = 0;
2
3 void setup(){
4   Serial.begin(9600);
5 }
6
7 void loop() {
8   Serial.println(analogRead(analogPin));
9   delay(100);
10 }
```

At the bottom of the IDE, there are three status bars. The top one says "Compiling sketch...". The middle one says "Uploading...". The bottom one says "Done uploading." and is highlighted with a green checkmark.

Compiling sketch...

Uploading...

Done uploading.



Check it!
Use the Serial
monitor to check if
your code is working.
Your monitor should
show numbers that
respond to our
sensor input!

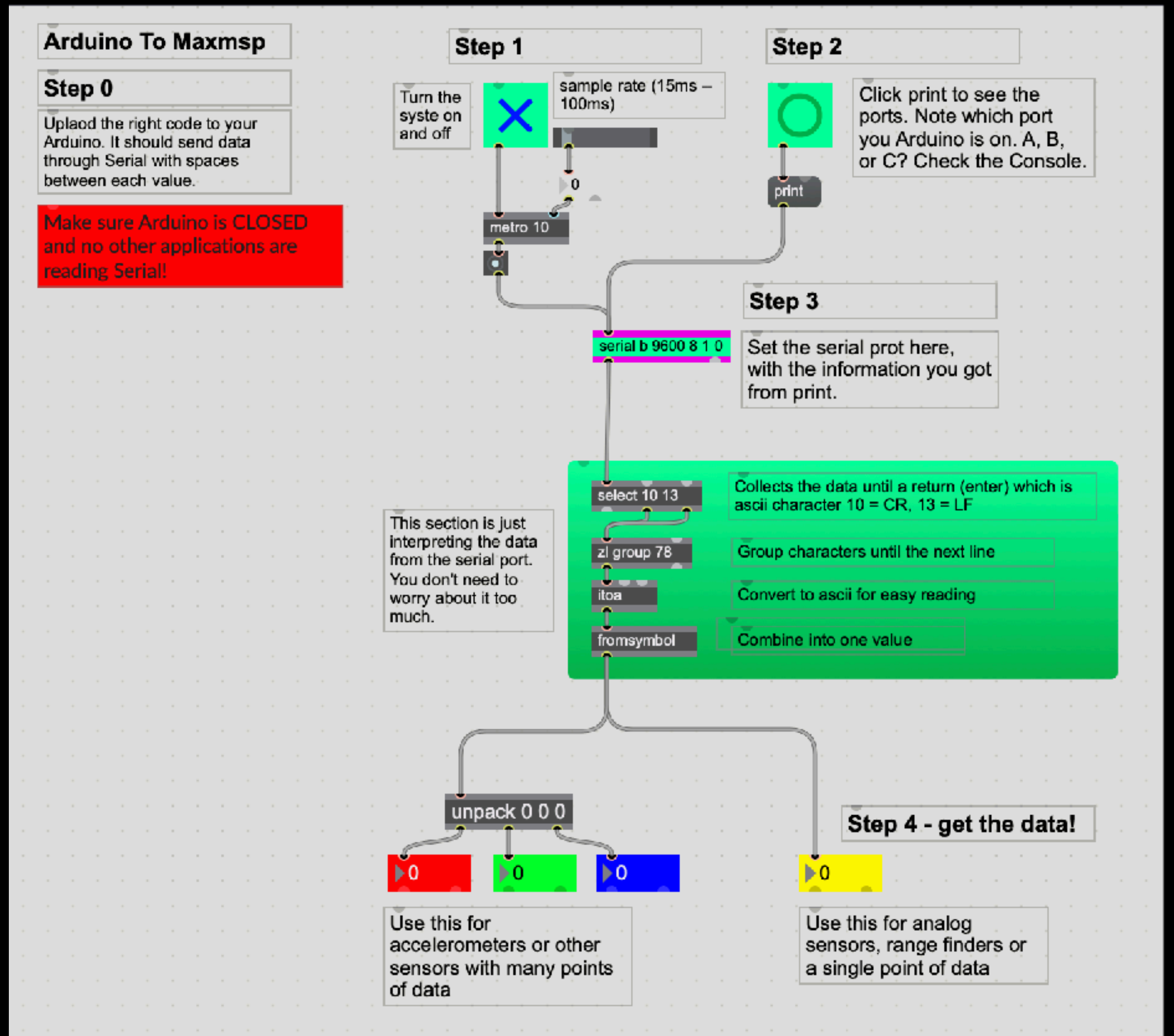


Now lets move to maxmsp.

Serial connections can only communicate to one program at a time. So make sure Arduino is closed.



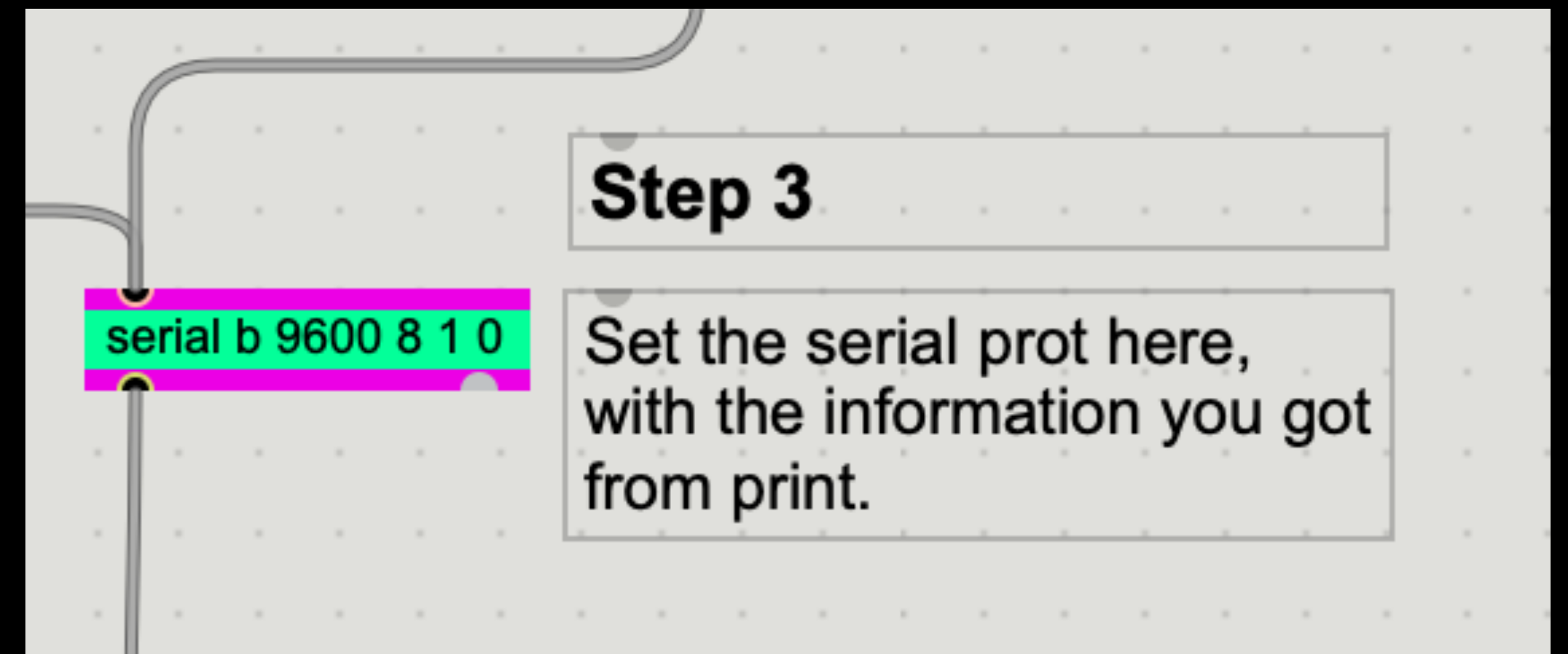
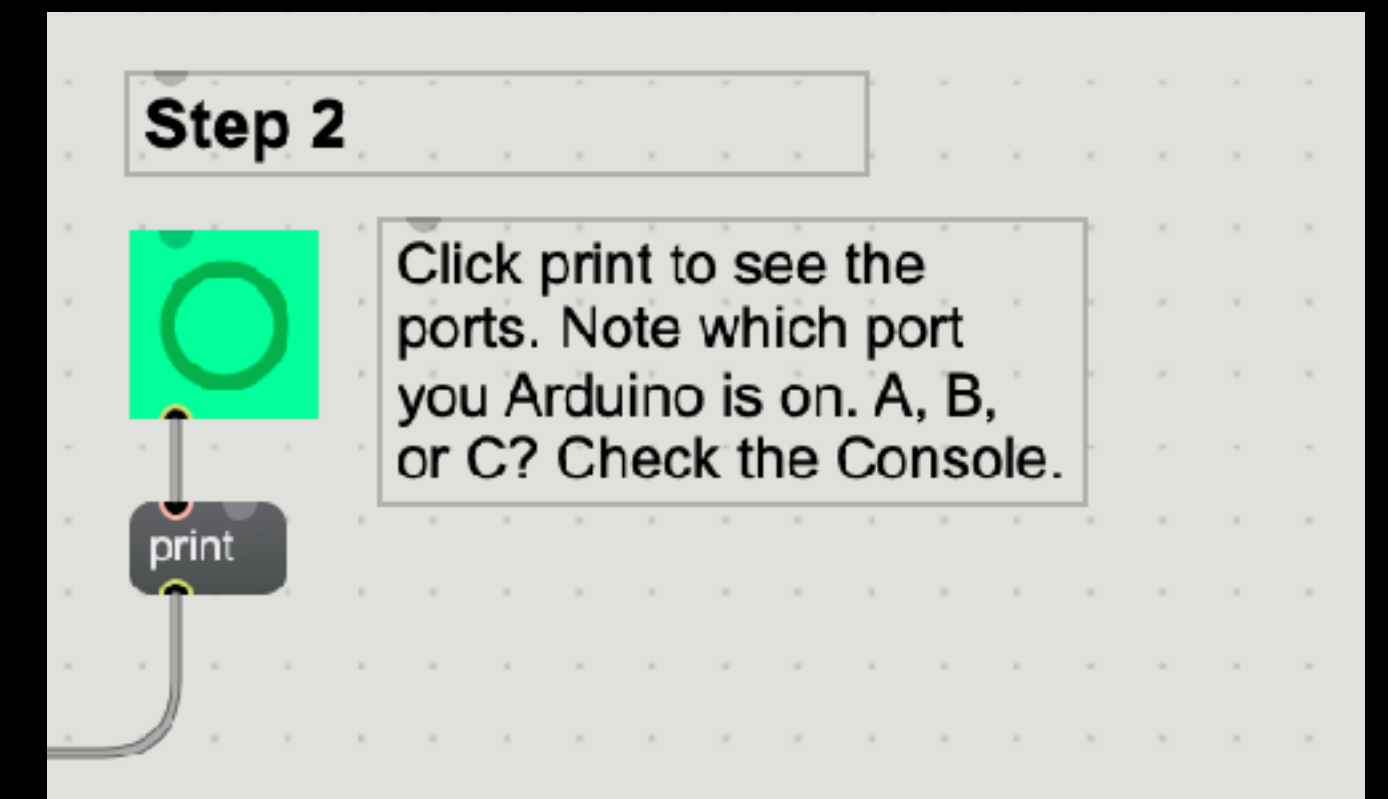
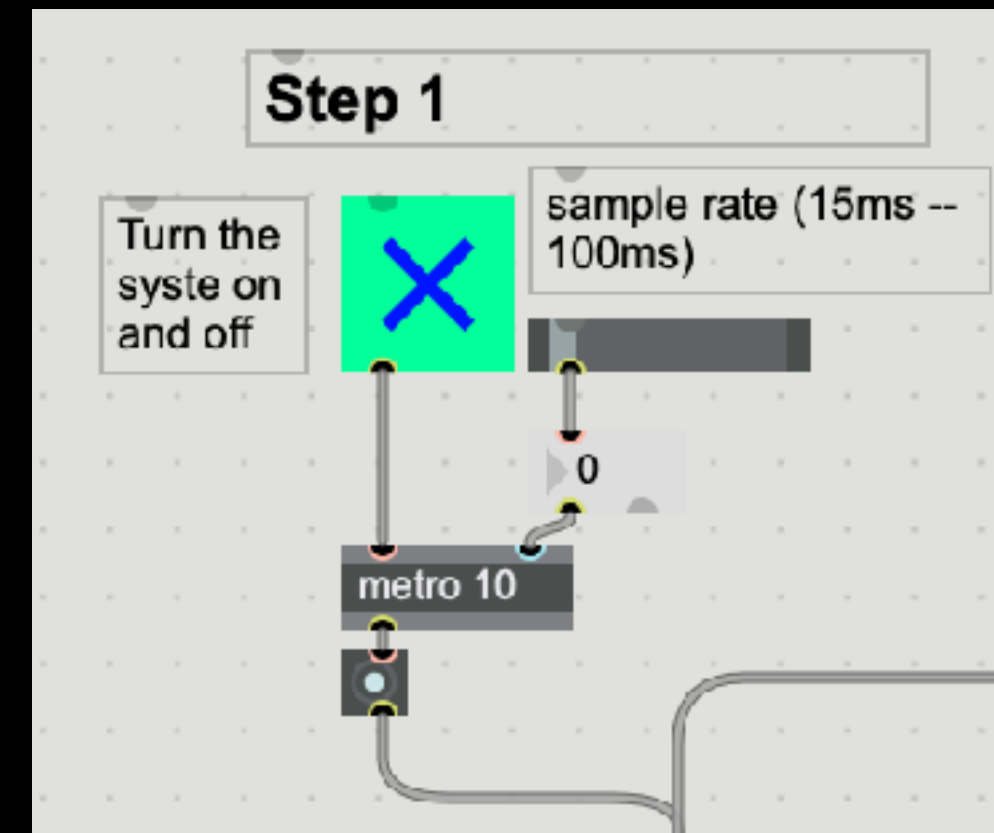
Arduino-To-Max patch Open arduino-to- max.maxpat



Arduino-To-Max patch
Set the metro to begin gathering data from the serial with a toggle.

Print all of the ports,
select the correct port

Enter the correct port
in the serial read object

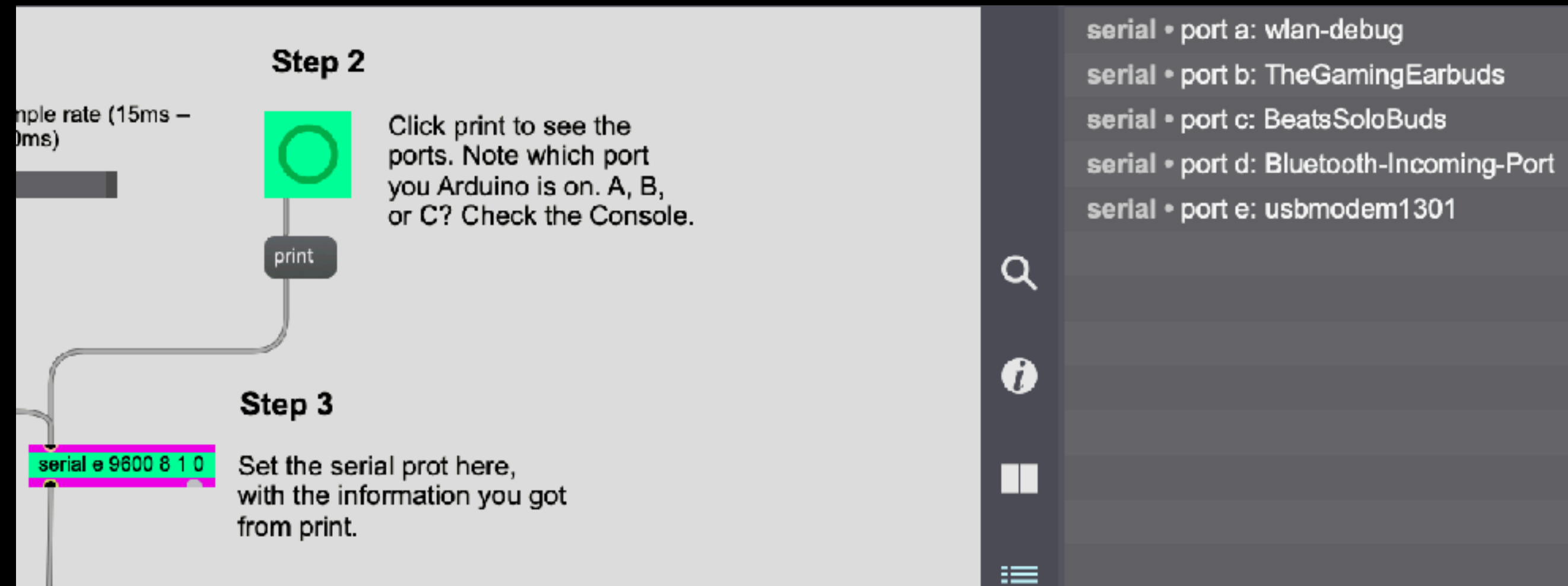


Arduino-To-Max patch

- Print will show all ports available in the **print log**.

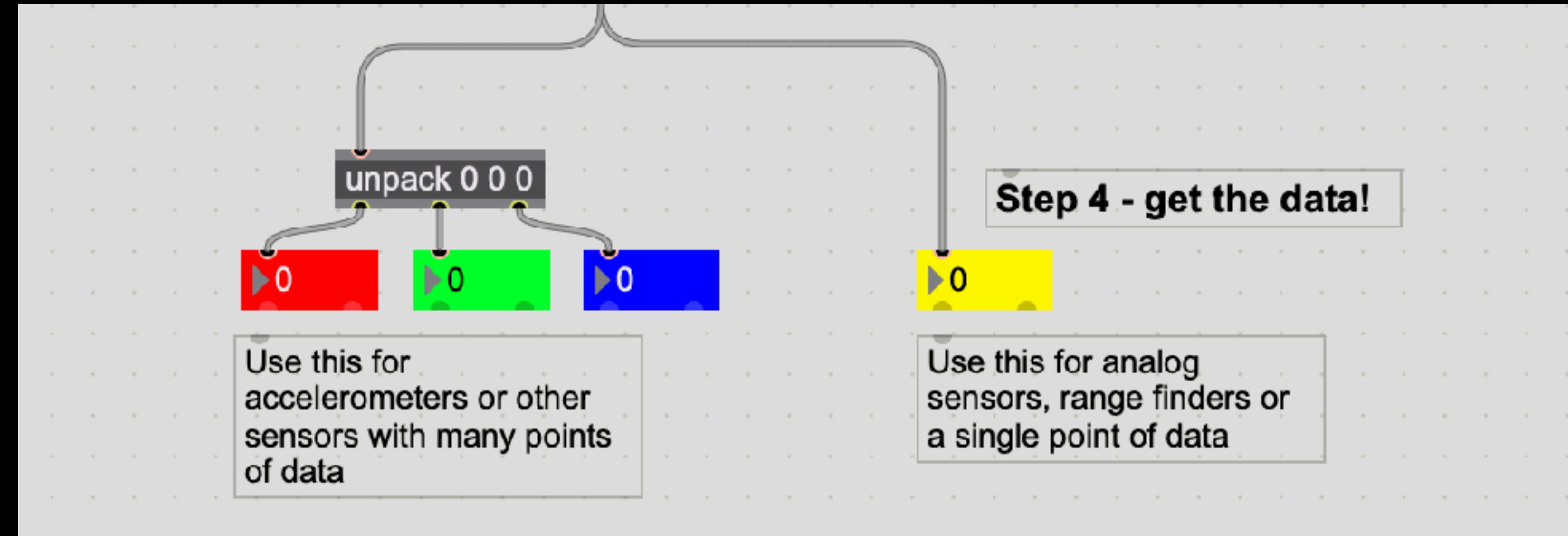
Usbmodem or Arduino Uno is usually the one we want, depending on the model you have.

- Add the port letter in the green box. Here it is e.



Arduino-To-Max patch

You can use the data by either unpacking a list of multiple values, or taking the single value. In the program we wrote together, there is only 1 value.



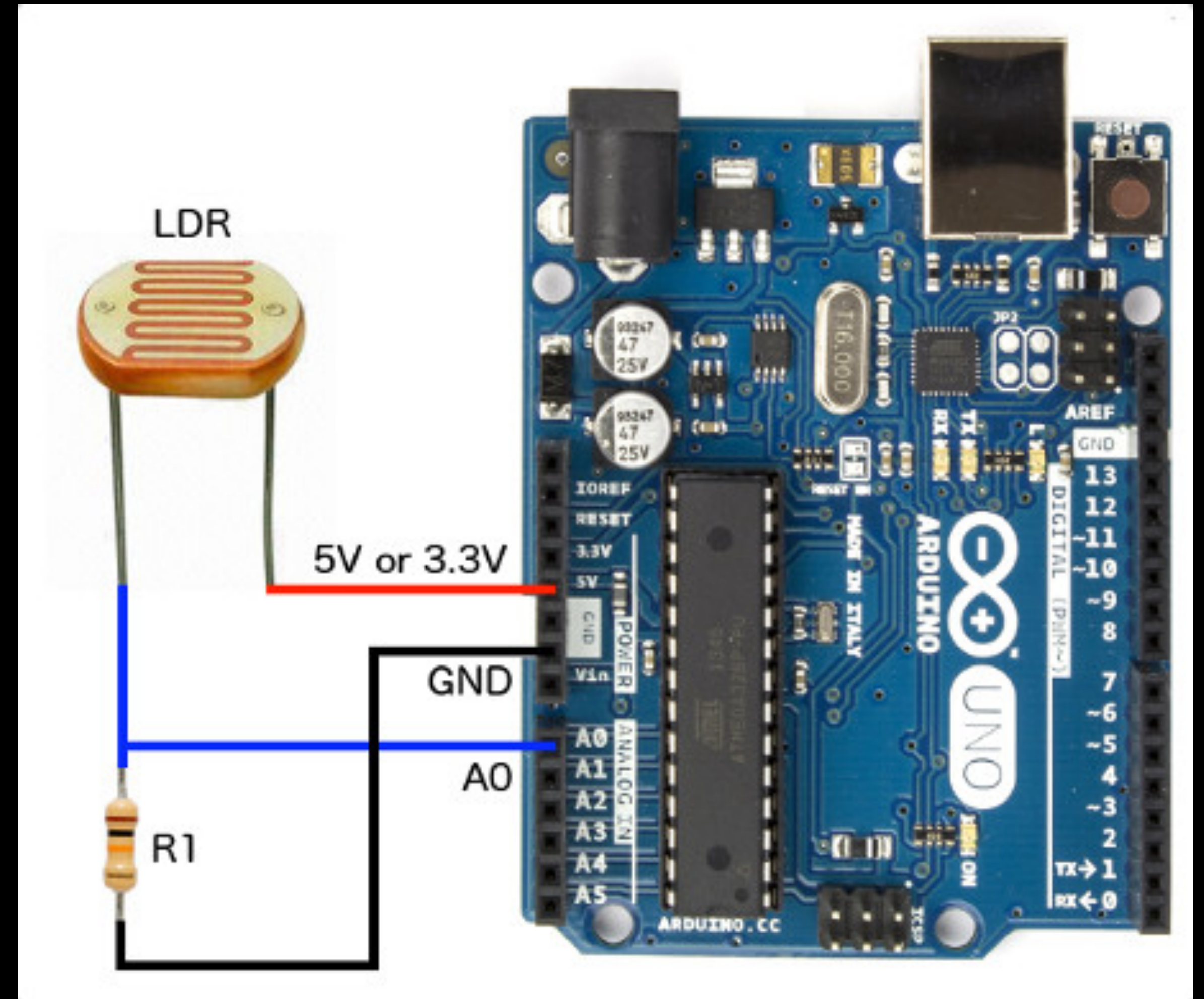
Try another sensor

- Pay attention to COLOR of wires in the instructions
- Read what is written on the board usually in white
- Take it one step at a time
- Double / triple check your work!

All analog sensors (Light, Flex, Pressure, knob)

- have 3 wires
- Red, Black, and (blue or yellow).
- Red > 5v
- Black > GND
- Yellow/Blue > A0

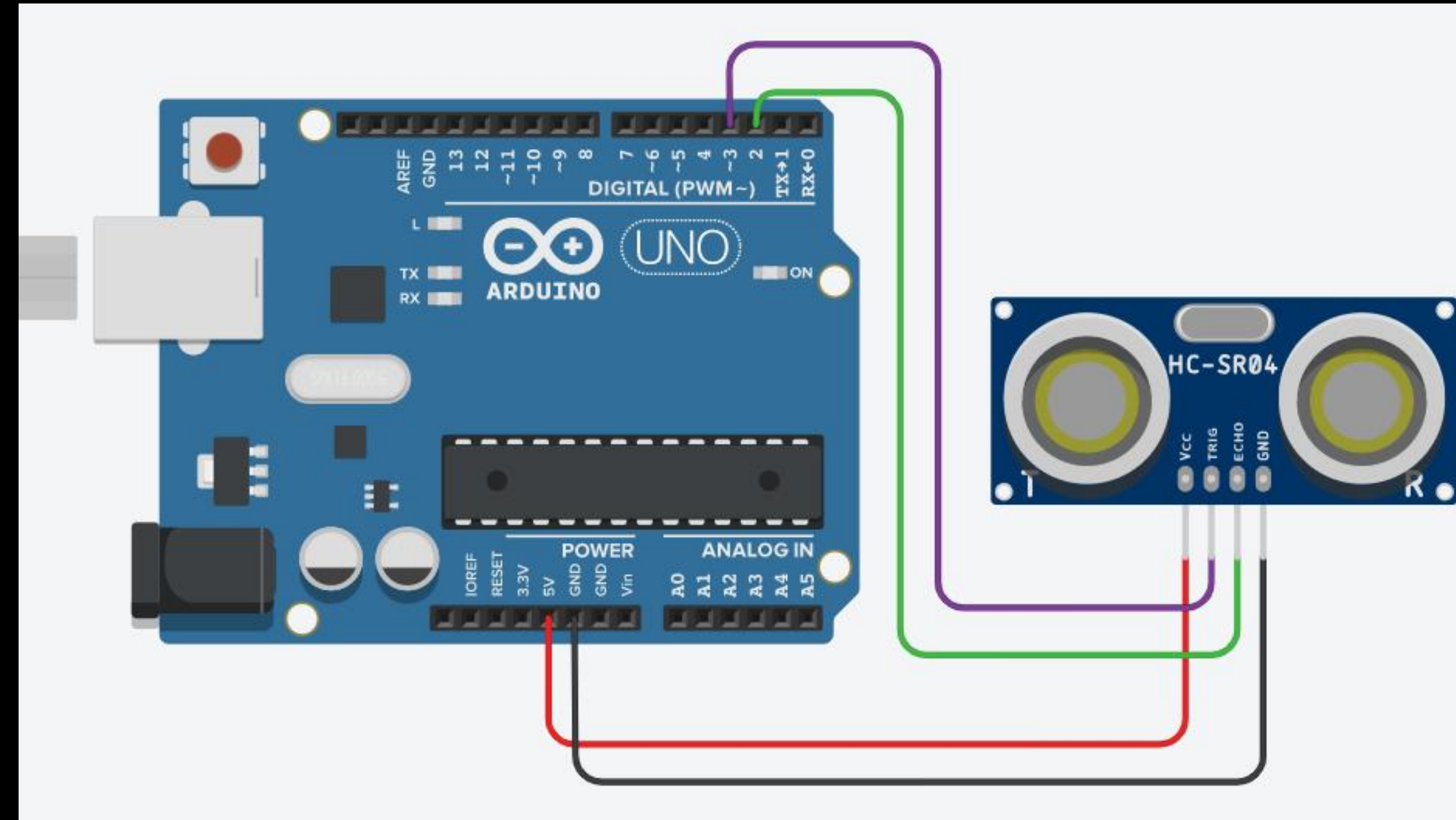
Output: You will get 1 number that indicates the value of the sensor



All analog sensors

- 4 wires
- Read the card writing
- GND > GND
- VCC > 5V
- TRIG > 3
- ECHO > 2

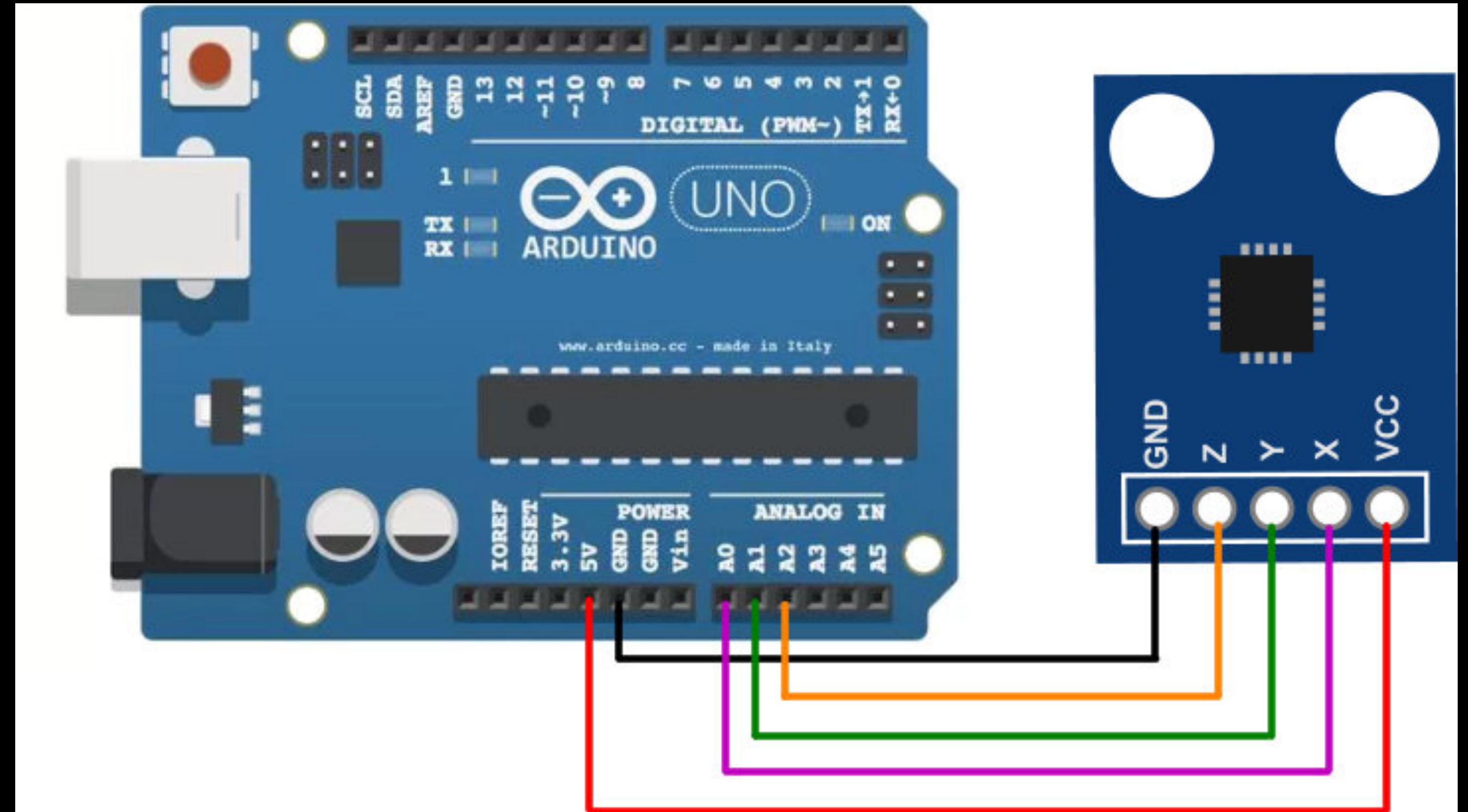
Output: you will get 1 number that indicates distance away from the sensor



All analog sensors

- 5 wires
- GND > GND
- VCC > 5V
- Z > A2
- Y > A1
- X > A0

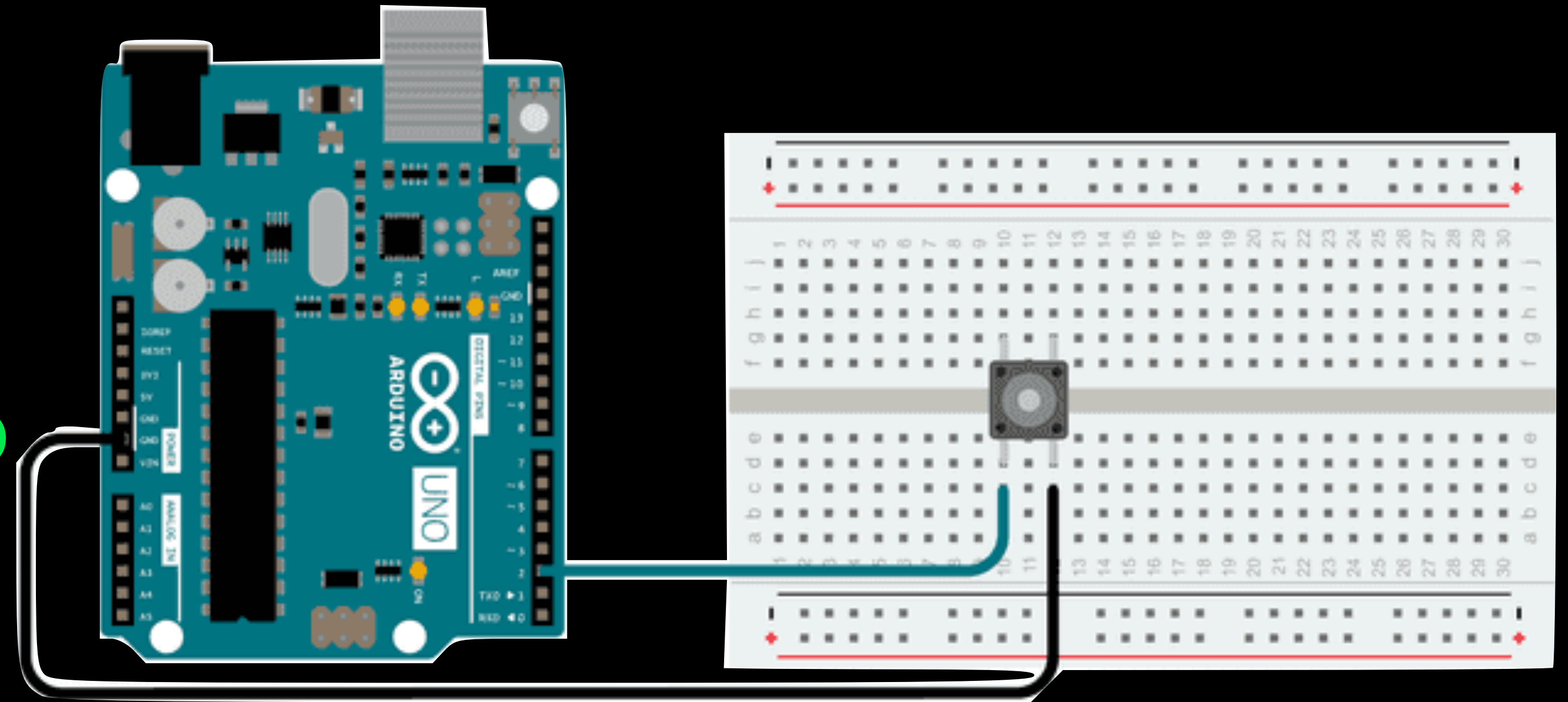
Output: You will get 3 numbers that correspond to the XYZ axis of rotation. Use the 3 number option in the maxpatch



Buttons

- GND
- 2

Buttons are just a broken connection, so feel free to use anything as a button!
More on this next week...



Output: You will get a 0 or 1 depending if the button is pressed or not

Code

Find all code here, or write your own!

[https://github.com/LeeCyborg/
Arduino-To-MaxMSP/tree/main](https://github.com/LeeCyborg/Arduino-To-MaxMSP/tree/main)

Download the Arduino IDE to load
code at [arduino.cc](https://www.arduino.cc)

Sensor Exploration - Due next week

- Work in pairs or teams
- Explore sensors in different contexts
- Integrate into patches, or modify the Arduino code (if you're up for it!)
- Get at least 1 sensor working and integrate it into something creative

Some things to explore:

- How can you integrate sensors into an environment?
- How can sensors interact with the viewer ?
- Consider that you can extend wires to virtually any length, integrate them into existing or fabricated

Sensors for your final:

- You can buy your own, borrow from me, talk to James to use the equipment available to you. Arduino's are available to borrow.
 - More advanced and specific sensors might need to be purchased by you
 - No need to use Arduino in your final.
- In-class work can be done for your sensor exploration project.

Homework:

- Start thinking about your final projects
(completely open project)

Next week:

**More sensor exploration with DIY sensors,
sensor exploration due!**